Departamento de Ingeniería civil y Agrícola
Facultad de Ingeniería

HOMEWORK 1

Introduction to probability and statistics and analysis of extreme events

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Important notes

- The computational processing must be made in the `R programming language`.
- The report must be written concisely in TEX, `markdown` (`R` report format) or `MS Word`. It must contain the plots and the analysis based on the results.
- The report must be named as `FirstSurnameSecondSurname_hwk1_SMWR_2025_II.pdf` (`MoralesMarin_hwk1_SMWR_II.pdf`). This must be sent to lmoralesm@unal.edu.co with the asunto `hwk1_SMWR`

1 Part I: Introduction to probability and statistics

1. From the *IDEAM*, *CAR*, *IDIGER* or other national or regional agencies, obtain a *time series (TS)* of *hourly precipitation* in *mm* in any place in Colombia. The series must have at least 15 yrs of continuous hourly records.
2. From the hourly TS, build a series of the following data: 1) duration (hours) of rainfall events, 2) intensity (mm of rainfall) of the continuous event and 3) time (hours) between rainfall events.
3. Accumulate the hourly TS to *daily*, *monthly* and *annual* precipitation.
4. Estimate the *mean*, the *variance*, the *skewness*, the *kurtosis* and the *interquartile range* for the original TS and the TS build in 2 and 3. Organize the results in tables and perform the analysis of the results.
5. For each of the TS in 2 and 3, make the following plots: the *series*, the *histogram* and *cumulative histogram.* Analyze and discuss the results.
6. For each of the series in 2 and 3: estimate the *pdf* and the *cdf* using at least two different methods (e.g. *method of moments* and *method of maximum likelihood*). Apply *goodness-of-fit tests* to choose the best *pdf*/ *cdf* that describe the data; made $Q-Q$ plots. Plot the chosen *pdf*s and *cdf*s and organize the distribution parameters in tables. Analyse and discuss the results.
7. Estimate the *confidence intervals* for the *mean* and for the *variance* of the TS in 2 and 3.
8. Estimate the join *pdf* between the *precipitation intensity* and *duration* and plot it. Estimate the *correlation* between both variables. Analyze and discuss the results.

2 Part II: Frequency analysis of extremes

1. Using the monthly precipitation TS, choose the maximum (P_{max}) and the minimum (P_{min}) annual precipitations.
2. From the observed data, estimate the *frequency curve* $P_{max/min}$ vs T (return period) for both time series.
3. Estimate the theoretical *frequency curves* for both TS by fitting a *pdf* /*cdf* to the data. Apply *goodness-of-fit tests* to choose the best *pdf*/ *cdf* that describe the data; made $Q-Q$ plots. Plot the chosen *pdf*s and *cdf*s and organize the distribution parameters in tables. Analyse and discuss the results.
4. Estimate the P_{max} and P_{min} for the following return periods: $T = 10, 20, \cdots, 200$ years. Estimate the confidence intervals for each P_{max} and P_{min}.